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PHYSICS

1) Find a vector having magnitude double of $3\hat{i} + 4\hat{j}$ and direction same as $5\hat{i} + 12\hat{j}$:-

- (1) $\frac{5\hat{i} + 12\hat{j}}{13}$
- (2) $\frac{10}{13}(3\hat{i} + 4\hat{j})$
- (3) $\frac{10}{13}(5\hat{i} + 12\hat{j})$
- (4) None

2) For two vectors $\vec{A}$ and $\vec{B}$, sum $(\vec{A} + \vec{B})$ is perpendicular to the difference $(\vec{A} - \vec{B})$. The ratio of their magnitudes is :-

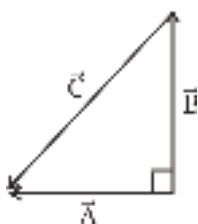
- (1) 1
- (2) 2
- (3) 3
- (4) None of these

3) The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ if the magnitude of $\vec{Q}$ is doubled, the new resultant becomes perpendicular to $\vec{P}$ then the magnitude of $\vec{R}$ is :-

- (1) $P + Q$
- (2) Q
- (3) P
- (4) $\frac{P + Q}{2}$

4) The resultant of $\vec{P}$ and $\vec{Q}$ is perpendicular to $\vec{P}$. What is the angle between $\vec{P}$ and $\vec{Q}$?

- (1) $\cos^{-1}(P/Q)$
- (2) $\cos^{-1}(-P/Q)$
- (3) $\sin^{-1}(P/Q)$
- (4) $\sin^{-1}(-P/Q)$



5) In a triangle, shown in figure :-

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- (1) $\vec{A} + \vec{B} = \vec{C}$
- (2) $\vec{A} + \vec{B} + \vec{C} = \vec{0}$
- (3) $\vec{B} + \vec{C} - \vec{A} = \vec{0}$
- (4) $\vec{A} + \vec{C} = \vec{B}$

6) Find the vector that must be added to the vector $\hat{i} - 3\hat{j} + 2\hat{k}$ and $3\hat{i} + 6\hat{j} - 7\hat{k}$ so that the resultant vector is a unit vector along the y-axis.

- (1) $-4\hat{i} - 2\hat{j} + 5\hat{k}$
- (2) $-4\hat{i} + 2\hat{j} + 5\hat{k}$
- (3) $4\hat{i} - 2\hat{j} + 5\hat{k}$
- (4) $-4\hat{i} - 2\hat{j} - 5\hat{k}$

7) A force of 6 N and another of 8 N can be applied together to produce the effect of a single force of :-

- (1) 1 N
- (2) 9 N
- (3) 15 N
- (4) 22 N

8) The angle between the vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be :-

- (1) 90°
- (2) 0°
- (3) 60°
- (4) 45°

9) If three vectors A, B and C are 12, 5 and 13 in magnitude such that $\vec{C} = \vec{A} + \vec{B}$, then the angle between A and B is

- (1) 60°
- (2) 90°
- (3) 120°
- (4) None

10) Five equal forces of 10 N each are applied at one point and all are lying in one plane. If the angles between them are equal, the resultant of these forces will be

- (1) Zero
- (2) 10N
- (3) 20N
- (4) $10\sqrt{2}$ N

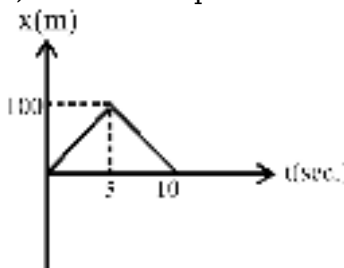
11) If $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be :-

- (1) 90°
- (2) 120°
- (3) 0°
- (4) 60°

12) If $|\vec{A}| = 2$ and $|\vec{B}| = \sqrt{3}$ and angle between $\vec{A}$ and $\vec{B}$ is 30° , then resultant will be :

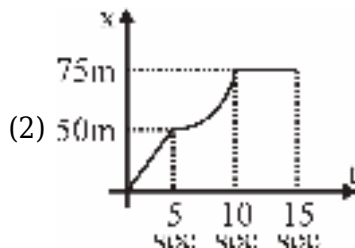
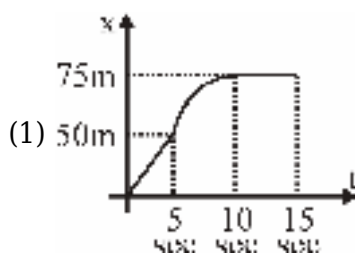
- (1) $\sqrt{13}$
- (2) $\sqrt{19}$
- (3) $\sqrt{12}$
- (4) $\sqrt{7}$

13) Find the displacement of the particle from following position-time graph:

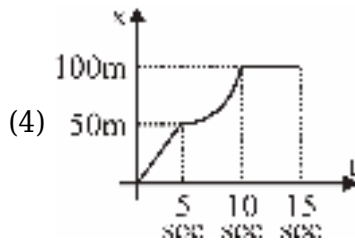
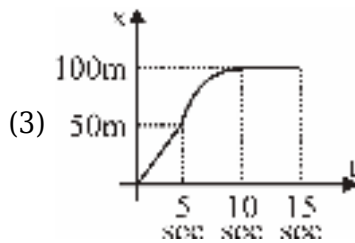


- (1) 200 m
- (2) 0 m
- (3) 10 m
- (4) 100 m

14) A car is moving with uniform velocity 10 m/s. After $t = 5$ sec, brakes are applied to produce constant retardation and car comes to rest at $t = 10$ sec. The position-time graph from $t = 0$ to 15 s is :



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15) A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})$ m/s, where $\hat{i}$ is along the horizontal and $\hat{j}$ is along the vertical. If $g = 10\text{m/s}^2$, the equation of its trajectory is :-

- (1) $y = x - 5x^2$
- (2) $y = 2x - 5x^2$
- (3) $4y = 2x - 5x^2$
- (4) $4y = 2x - 25x^2$

16) **Assertion :-** A body can have acceleration even if its velocity is zero at a given instant of time.

Reason :- A body is momentarily at rest when it reverse its direction while moving in a straight line.

- (1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are true but R is not a correct explanation of the Assertion.
- (3) A is true but R is false.
- (4) Both Assertion and Reason are false.

17) A body falling from height 'h' takes t_1 time to reach the ground. The time taken to cover the first $\left(\frac{1}{4}\right)^{\text{th}}$ of the height is :

(1) $\frac{t_1}{\sqrt{2}}$

(2) $\frac{t_2}{\sqrt{2}}$

(3) $\frac{t_1}{2}$

(4) None of these

18) A ball is thrown vertically upwards from the top of a tower with velocity 4.9 ms^{-1} . It strikes the pond near the base of the tower after 3 seconds. The height of the tower is :-

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- (1) 73.5 m
- (2) 44.1 m
- (3) 29.4 m
- (4) None of these

19) Two balls are dropped from heights h and $2h$ respectively from the earth surface. The ratio of time of these balls to reach the earth is:-

- (1) $1 : \sqrt{2}$
- (2) $\sqrt{2} : 1$
- (3) $2 : 1$
- (4) $1 : 4$

20) A stone is released from the top of a tower. If its velocity at half of the height is 10 m/s , then height of the tower is: ($g = 10 \text{ ms}^{-2}$)

- (1) 8 m
- (2) 10 m
- (3) 12 m
- (4) 16 m

21) A balloon is going vertically upwards with a velocity of 10 m/s . When it is 75 m above the ground, a stone is gently released from it. The time taken by the stone to reach the ground is ($g = 10 \text{ m/s}^2$)

- (1) 3 s
- (2) 4 s
- (3) 5 s
- (4) 6 s

22) A rocket is fired upward from the earth's surface such that it creates an acceleration of 20 m/sec^2 . If after 5 sec its engine is switched off, the maximum height of the rocket from earth's surface would be :- ($g = 10 \text{ m/s}^2$)

- (1) 250 m
- (2) 500 m
- (3) 750 m
- (4) 1000 m

23) A ball is projected vertically upward from ground at 20 m/s . Immediately after 3 sec what will be the direction of displacement, velocity and acceleration ? ($g = 10 \text{ m/s}^2$)

	Displacement	Velocity	Acceleration
(1)	up	down	down
(2)	down	down	down
(3)	up	down	up
(4)	up	up	down

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- (1) 1
- (2) 2
- (3) 3
- (4) 4

24) A ball is thrown vertically upwards with velocity 600 m/s. Calculate distance travelled in last 2 sec of its upward motion :-

- (1) 20 m
- (2) 30 m
- (3) 10 m
- (4) 25 m

25) A stone is dropped into a well in which the level of water is h below the top of the well. If v is velocity of sound, the time T after which the splash is heard is given by.

- (1) $T = \frac{2h}{v}$
- (2) $T = \sqrt{\frac{2h}{g}} + \frac{h}{v}$
- (3) $T = \sqrt{\frac{2h}{v}} + \frac{h}{g}$
- (4) $T = \sqrt{\frac{h}{2g}} + \frac{2h}{v}$

26) A ball is dropped from certain height such that the distance travelled in last second is equal to the distance travelled in first three seconds. Height from which the ball is dropped, is

- (1) 80 m
- (2) 180 m
- (3) 145 m
- (4) 125 m

27) From a tower of height H a body is projected vertically upward with speed ' u '. It strikes the ground with speed $4u$. The total time of flight is

- (1) $\frac{3u}{g}$
- (2) $\frac{5u}{g}$
- (3) $\frac{2u}{g}$
- (4) $\frac{u}{g}$

28) A body is thrown upwards and reaches maximum height. At that position its :-

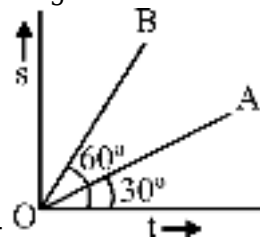
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- (1) Velocity is zero and acceleration is also zero
- (2) Velocity is zero but acceleration is maximum
- (3) acceleration is minimum
- (4) Velocity is zero and acceleration is equal to g .

29) A ball thrown vertically upwards with a speed of 20 m/s from the top of a tower returns to the earth in 6 sec. Find the height of the tower ? ($g = 10 \text{ m/s}^2$)

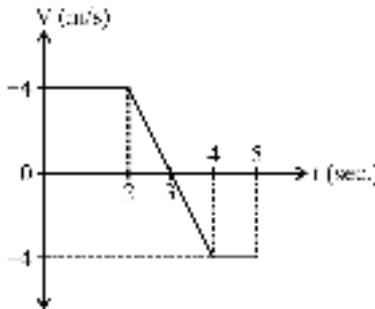
- (1) 40 m
- (2) 60 m
- (3) 80 m
- (4) 100 m

30) The displacement-time graph for two particles A and B are straight lines inclined at angles of



30° and 60° with the time axis. The ratio of velocity of $V_A : V_B$ is :-

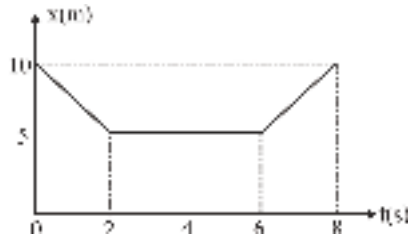
- (1) 1 : 2
- (2) $1 : \sqrt{3}$
- (3) $\sqrt{3} : 1$
- (4) 1 : 3



31) The velocity-time graph of a particle moving along straight line is shown in figure. Find the displacement of the particle in 5 seconds :-

- (1) 16 m
- (2) 4 m
- (3) 12 m
- (4) 10 m

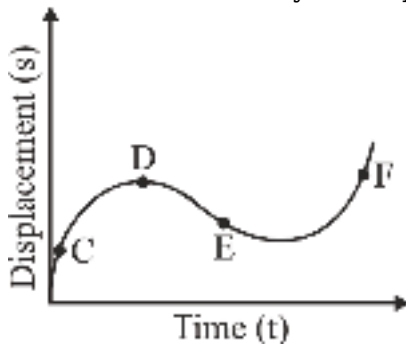
32) Figure shows the position time graph for a particle in one dimensional motion. What is the



displacement of the particle in first 4 seconds:-

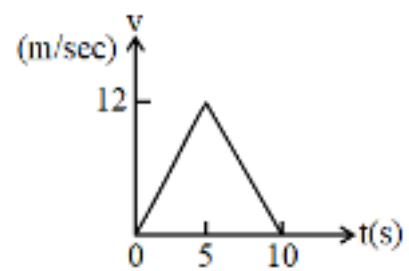
- (1) 5 m
- (2) -5 m
- (3) 0
- (4) 15 m

33) The displacement-time graph of a moving particle is shown in the adjoining figure. The instantaneous velocity of the particle is negative at the point :-



- (1) D
- (2) F
- (3) C
- (4) E

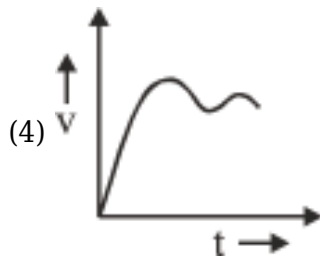
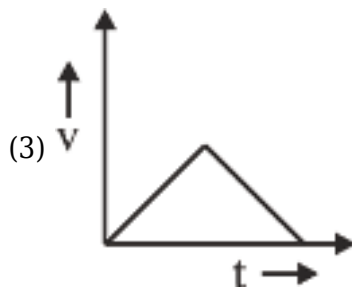
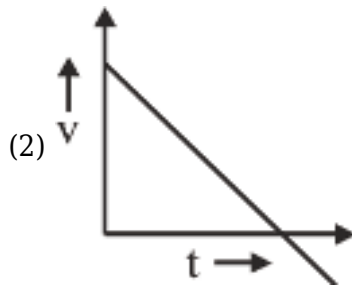
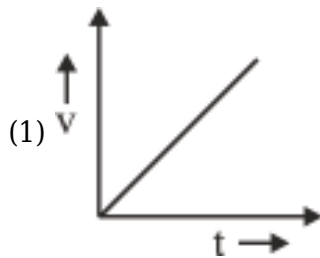
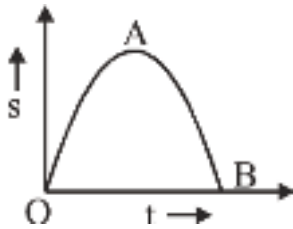
34) The speed-time graph of a particle moving along a fixed direction as shown in the figure. The



distance transversed by the particle between $t = 0$ s to $t = 10$ s is :

- (1) 20 m
- (2) 40 m
- (3) 60 m
- (4) 80 m

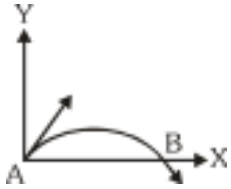
35) Velocity-time graph corresponding to displacement-time graph shown in adjoining figure is :-



36) A missile is fired for maximum range with an initial velocity of 20 m/s. If $g = 10 \text{ m/s}^2$, the range of the missile is :-

- (1) 40 m
- (2) 50 m
- (3) 60 m
- (4) 20 m

37) The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j})$ m/s. Its velocity (in m/s) at point B is :-

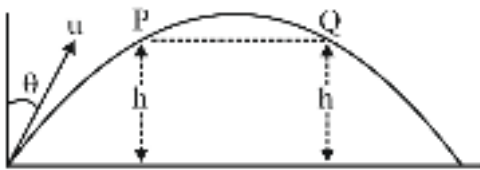


- (1) $2\hat{i} + 3\hat{j}$
- (2) $-2\hat{i} - 3\hat{j}$
- (3) $-2\hat{i} + 3\hat{j}$
- (4) $2\hat{i} - 3\hat{j}$

38) The range of a projectile projected at an angle 15° with horizontal is $100\sqrt{3}\text{m}$. It is fired with the same speed at an angle of 60° with vertical, its range will be

- (1) $100\sqrt{3}\text{m}$
- (2) 100 m
- (3) 300 m
- (4) $300\sqrt{3}\text{m}$

39) A particle is thrown with velocity u making an angle θ with the vertical. It just crosses the top of two poles each of height h after 1 s and 3 s respectively. The maximum height of projectile is $[g = 9.8 \text{ m/s}^2]$ –



- (1) 9.8 m
- (2) 19.6 m
- (3) 39.2 m
- (4) 4.9 m

40) A body is projected such that the horizontal range is three times the greatest height. The angle of projection is:

- (1) 25°
- (2) 33°
- (3) 42°
- (4) 53°

41) A particle is projected at an angle of 45° , 6m away from the foot of a wall, just touches the top of the wall and falls on the ground on the opposite side at a distance 3m from it. The height of wall is

- (1) 2m
- (2) $\frac{4}{3}\text{m}$

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(3) $\frac{8}{3}m$

(4) $\frac{3}{4}m$

42) Two stones are projected horizontally from the same height with speeds 100 m/s and 40 m/s. The ratio of their horizontal range is

(1) 1 : 1

(2) 5 : 2

(3) 2 : 5

(4) 3 : 4

43)

An aeroplane moving horizontally with a speed of 180 km/h drops a food packet while flying at a height of 490 m. The horizontal range of the packet is :

(1) 180 m

(2) 980 m

(3) 500 m

(4) 670 m

44)

Two paper screens A and B are separated by distance 100 m. A bullet penetrates A and B, at points P and Q respectively, where Q is 10 cm below P. If bullet is travelling horizontally at the time of hitting A, the velocity of bullet at A is nearly

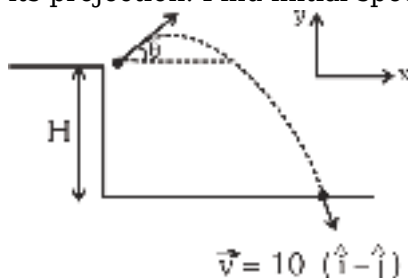
(1) 100 m/s

(2) 200 m/s

(3) 600 m/s

(4) 700 m/s

45) A ball was thrown from height H and the ball hit the floor with velocity $10(\hat{i} - \hat{j})$ after 1.5 sec of its projection. Find initial speed of ball.



(1) $10\sqrt{5}$ m/s

(2) $5\sqrt{5}$ m/s

(3) 15 m/s

(4) 30 m/s

CHEMISTRY

1)

200 gm CaCO_3 heated and it decomposed into CaO and CO_2 . What is the weight of residue -(Atomic weight of $\text{Ca} = 40$)

- (1) 56 gm
- (2) 200 gm
- (3) 100 gm
- (4) 112 gm

2) What will be molarity of 20%(w/v) of NaOH ?

- (1) 5
- (2) 0.5
- (3) 20
- (4) 2

3)

3.011×10^{22} atoms of an element weight 1.15 gm. The atomic mass of the element is:-

- (1) 10 a.m.u
- (2) 2.3 a.m.u
- (3) 35.5 a.m.u
- (4) 23 a.m.u.

4) Match the following prefixes with their multiples :-

(a)	Micro	(i)	10^{-12}
(b)	Mega	(ii)	10^{-6}
(c)	Giga	(iii)	10^6
(d)	Pico	(iv)	10^9

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

5) How many electrons are present in 100 g of calcium carbonate?

- (1) 1.505×10^{24}
- (2) 2.047×10^{24}
- (3) 3.011×10^{24}

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(4) 3.011×10^{25}

6) Volume of 2 mole of gas occupied at 2 atm and 273 K is :-

- (1) 22.4 Litre
- (2) 44.8 Litre
- (3) 11.2 Litre
- (4) 67.2 Litre

7)

Number of neutrons present in 18 g of water is:-

- (1) $10 N_A$
- (2) $0.8 N_A$
- (3) $0.3 N_A$
- (4) $8 N_A$

8) The vapour densities of two gases are in the ratio of 2:3. Their molecular masses are in the ratio of

- (1) 3:2
- (2) 6:1
- (3) 2:3
- (4) 1:6

9) A compound used in making nylon, contains 43.8% oxygen. There are four oxygen atoms per molecule. What is the molecular mass of compound ?

- (1) 36
- (2) 116
- (3) 292
- (4) 146

10) **Assertion :** Equal moles of different substances contain same number of constituent molecules.

Reason : Equal weights of different substance contain the same number of constituent molecules.

- (1) Both Assertion and Reason are correct and the Reason is the correct explanation of Assertion
- (2) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion
- (3) Assertion is correct but Reason is incorrect
- (4) Both the Assertion and Reason are incorrect

11) CaCO_3 is 90% pure. Volume of CO_2 collected at STP when 10 g of CaCO_3 is decomposed is-

- (1) 2.016 L
- (2) 1.008 L

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(3) 10.08 L

(4) 20.16 L

12) A cylinder provided with piston contains PCl_5 , PCl_3 and Cl_2 at equilibrium. If the system is compressed with the help of piston. Then choose the correct statement.

(1) Some more PCl_5 will be decomposed.

(2) System remains unaffected

(3) PCl_3 and Cl_2 react to form PCl_5

(4) Explosion takes place

13) For an equilibrium $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$ which of the following statement is true ?

(1) The pressure changes don't affect the equilibrium

(2) More of ice melts if pressure on the system is increased

(3) More of the liquid freezes if pressure on the system is increased

(4) Melting point of ice increases

14) The gaseous reaction $\text{A} + \text{B} \rightleftharpoons 2\text{C} + \text{D} + \text{Q KJ}$ is most favoured at :

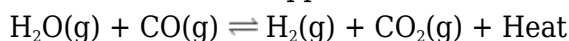
(1) Low temperature and high pressure

(2) High temperature and high pressure

(3) High temperature and low pressure

(4) Low temperature and low pressure

15) If the pressure in a reaction vessel for the following reaction is increased by decreasing the volume, what will happen to the concentrations of CO and CO_2 ?



(1) both the $[\text{CO}]$ and $[\text{CO}_2]$ will decrease

(2) neither the $[\text{CO}]$ nor the $[\text{CO}_2]$ will change

(3) the $[\text{CO}]$ will decrease and the $[\text{CO}_2]$ will increase

(4) both the $[\text{CO}]$ and $[\text{CO}_2]$ will increase

16) **Assertion:-** Addition of catalyst to a reaction mixture, increases the amount of reactants and products at equilibrium.

Reason:- The forward reaction becomes slower on adding catalyst.

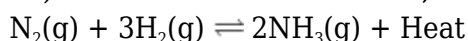
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True and the Reason is False.

(4) Both Assertion & Reason are False.

17) For the reversible reaction,



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the equilibrium shifts in forward direction :-

- (1) By increasing the concentration of $\text{NH}_3(\text{g})$
- (2) By decreasing the pressure
- (3) By decreasing the concentrations of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$
- (4) By increasing pressure and decreasing temperature

18) An aqueous solution of CH_3COONa will be

- (1) Acidic
- (2) Alkaline
- (3) Neutral
- (4) None of these

19) Calculate the % of hydrolysis in 0.034 M aqueous solution of KOCN (K_a of $\text{HOCN} = 3.33 \times 10^{-4}$ M)

- (1) 0.003%
- (2) 0.002%
- (3) 0.001%
- (4) 0.008%

20) Correct order of pH of following solution is ?

- (1) 0.1 M HCl
- (2) 0.1 M CH_3COOH
- (3) 0.1 M NaCl
- (4) 0.1 M KCN

- (1) (1) > (2) > (3) > (4)
- (2) (4) > (3) > (2) > (1)
- (3) (3) > (2) > (4) > (1)
- (4) (2) > (3) > (1) > (4)

21) An aqueous solution at room temperature is 0.1M with respect to ammonium chloride and 0.01M with respect to NH_4OH ($\text{p}K_b = 5$), the pH of the solution is :

- (1) 7.5
- (2) 6.8
- (3) 6.5
- (4) 8.0

22) Which of the following is a buffer solution ?

- (1) 500mL of 0.1 N CH_3COOH + 500 mL of 0.1N NaOH
- (2) 500mL of 0.1 N CH_3COOH + 500 mL of 0.1N HCl

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(3) 500 mL of 0.1 N CH_3COOH + 500 mL of 0.2N NaOH

(4) 500 mL of 0.2 N CH_3COOH + 500 mL to 0.1N NaOH

23)

Calculate the pH of a 10^{-3}M solution of $\text{Ba}(\text{OH})_2$ if it undergoes complete ionisation. ($K_w = 1 \times 10^{-14}$) :-

(1) 12.30

(2) 11.30

(3) 10.00

(4) 9.00

24) An acidic buffer is prepared by mixing :

(1) weak acid and it's salt with strong base

(2) equal volumes of equimolar solutions of weak acid and weak base

(3) strong acid and it's salt with strong base

(4) strong acid and it's salt with weak base (The pK_a of acid = pK_b of the base)

25) NH_4CN is a salt of weak acid HCN ($K_a = 6.2 \times 10^{-10}$) and a weak base NH_4OH ($K_b = 1.8 \times 10^{-5}$). 1 molar solution of NH_4CN will be :-

(1) Neutral

(2) Strongly acidic

(3) Weakly acidic

(4) Weakly basic

26) 0.01 M HA (aq.) is 2% ionized; $[\text{OH}^-]$ of solution is :-

(1) 2×10^{-4} M

(2) 10^{-8} M

(3) 5×10^{-11} M

(4) 2×10^{-12} M

27) The conjugated base of $(\text{CH}_3)_2\overset{+}{\text{N}}\text{H}_2$ is :-

(1) CH_3NH_2

(2) $(\text{CH}_3)_2\text{N}^+$

(3) $(\text{CH}_3)_2\text{N}$

(4) $(\text{CH}_3)_2\text{NH}$

28)

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In the reaction $\text{HNO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{NO}_3^-$, the conjugate base of HNO_3 is :-

- (1) H_2O
- (2) H_3O^+
- (3) NO_3^-
- (4) H_3O^+ and NO_3^-

29) Which of the following salts undergoes hydrolysis in water:-

- (1) Na_3PO_4
- (2) CH_3COONa
- (3) NaNO_3
- (4) Both of (1) and (2)

30) The pH of a solution is 2. Its pH is to be changed to 4. Then the H^+ concentration of original solution has to be:-

- (1) Halved
- (2) Doubled
- (3) Increased 100 times
- (4) Decreased 100 times

31) Which of the following salts will give highest pH in water ?

- (1) KCl
- (2) NaCl
- (3) Na_2CO_3
- (4) CuSO_4

32) The pH of solution A, B, C, D are 9.5, 2.5, 3.5 and 5.5 respectively. The most acidic solution is :-

- (1) A
- (2) B
- (3) C
- (4) D

33) What is the hydronium ion concentration of a 0.25 M HA solution? ($K_a = 4 \times 10^{-8}$) :-

- (1) 10^{-4} M
- (2) 10^{-5} M
- (3) 10^{-7} M
- (4) 10^{-10} M

34) **Statement-1** : pH of hydrochloric acid solution is less than that of acetic acid solution of the same concentration.

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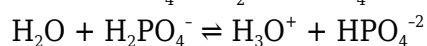
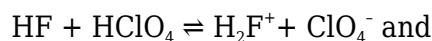
Statement-2 : In equimolar solutions, the number of protons present in hydrochloric acid is less than that present in acetic acid.

- (1) Statement 1 is True, Statement 2 is True; Statement 2 is a correct explanation of Statement 1.
- (2) Statement 1 is True, Statement 2 is True; Statement 2 is not a correct explanation for Statement 1.
- (3) Statement 1 is False, Statement 2 is True.
- (4) Statement 1 is True, Statement 2 is False.

35) What is the percent dissociation of 0.1 M solution of acetic acid ? ($K_a = 10^{-5}$)

- (1) 10%
- (2) 100%
- (3) 1%
- (4) 0.01%

36)



Select out the group of acidic species lying on the right side of the equilibrium :-

- (1) H_2F^+ , HPO_4^{2-}
- (2) H_3O^+ , H_2F^+
- (3) ClO_4^- , H_3O^+
- (4) ClO_4^- , HPO_4^{2-}

37) How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ?

- (1) 10 L
- (2) 0.9 L
- (3) 2.0 L
- (4) 9.0 L

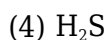
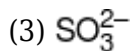
38) Which of the following solutions will have pH = 10 at 298 K?

- (1) 1×10^{-10} M HCl solution
- (2) 1×10^{-4} M NaOH solution
- (3) 1×10^{-10} M NaOH solution
- (4) Both (1) and (2)

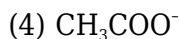
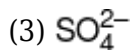
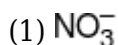
39) The conjugate base of HSO_3^- is :-

- (1) H_2SO_3
- (2) SO_2

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40) Strongest conjugate base among the following is :-



41) At infinite dilution a weak electrolyte contains :-

(1) Unionised molecules

(2) Only ions

(3) Ions and unionised molecules

(4) Not definite

42) Half of the NH_4OH is neutralised on addition of a HCl solution to it. If $K_b(\text{NH}_4\text{OH}) = 2 \times 10^{-4}$. Then pH of the solution is :-

(1) 9.2

(2) 5.7

(3) 7

(4) 10.3

43) 100 ml of 0.2 M H_2SO_4 is added to 100 ml of 0.2 M NaOH . The resulting solution will be :-

(1) Acidic

(2) Basic

(3) Neutral

(4) Slightly basic

44) pH of 10^{-5} M HCl ?

(1) 5

(2) 8

(3) 6.97

(4) 7

45) For the reaction $\text{AB}_{(g)} \rightleftharpoons \text{A}_{(g)} + \text{B}_{(g)}$, AB is 33% dissociated at a total pressure of P. Therefore, P is related to K_p by one of the following options:

(1) $P = K_p$

(2) $P = 3K_p$

(3) $P = 4K_p$

(4) $P = 8K_p$

BIOLOGY

1) Pyrenoids are made up of -

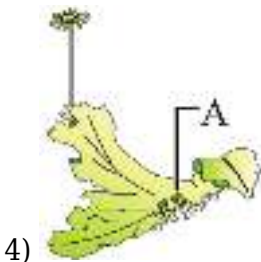
- (1) Core of starch surrounded by sheath of protein
- (2) Core of protein surrounded by fatty sheath
- (3) Proteinaceous centre and starchy sheath
- (4) Core of nucleic acid surrounded by protein

2) Green algae which is used as food supplement is :-

- (1) *Spirulina*
- (2) *Porphyra*
- (3) *Chlorella*
- (4) (1) & (3) both

3) Sexual reproduction oogamous type and accompanied by complex post fertilization development, can be observed in :-

- (1) Chlorophyceae
- (2) Phaeophyceae
- (3) Xanthophyceae
- (4) Rhodophyceae



Identify the plant and labelled structure "A".

- (1) Marchantia and Gemma cup for asexual reproduction
- (2) Riccia and Rhizoids
- (3) Marchantia and gemmae for sexual reproduction
- (4) Funaria and Scales

5) Read the following statements.

- (A) Presence chlorophyll a, d and r-Phycocyanin
- (B) Gametes are without flagella.
- (C) Stored energy in form of floridean starch.

All these information are related to :-

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- (1) Phaeophyceae
- (2) Chlorophyceae
- (3) Rhodophyceae
- (4) Cyanophyceae

6) In which of the following, all listed genera belong to the same class of algae ?

- (1) *Chara, Fucus, Polysiphonia*
- (2) *Volvox, Spirogyra, Chara*
- (3) *Porphyra, Ectocarpus, Ulothrix*
- (4) *Sargassum, Laminaria, Gracillaria*

7) Which of the following is living fossil ?

- (1) *Cedrus*
- (2) *Pinus*
- (3) *Ginkgo biloba*
- (4) All of these

8) Independent alternation of generation is the unique character of the life cycle of :-

- (1) Pteridophyta
- (2) Bryophyta
- (3) Gymnosperm
- (4) Angiosperm

9) Gametophyte of pteridophyte is :-

- (1) Small but multicellular, free living, mostly photosynthetic, differentiated into root, stem and leaf
- (2) Small, inconspicuous but multicellular dependent mostly photosynthetic thalloid body
- (3) Small but multicellular, free living mostly photosynthetic thalloid structure
- (4) Small, inconspicuous but multicellular free - living mostly non -photosynthetic thalloid body

10) In which plant, the female gametophyte is retained on the parent sporophyte for some periods ?

- (1) Fern
- (2) Cycas
- (3) Salvinia
- (4) Equisetum

11) How many structures are diploid in *Funaria* plant?

Rhizoids, Protonema, Spores, Foot, Capsule

- (1) 3
- (2) 2
- (3) 4

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(4) 6

12) In Phaeophyceae, the vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of :-

- (1) Algin
- (2) Carrageen
- (3) Agar
- (4) Starch



13) Identify the diagram given below and mention the class to which this plant belongs :-

- (1) *Equisetum*, Sphenopsida
- (2) *Dryopteris*, Psilopsida
- (3) *Funaria*, Hepaticopsida
- (4) *Funaria*, Bryopsida

14) When moss spores germinate, they form ?

- (1) Leaf gametophyte
- (2) sporophyte
- (3) prothallus
- (4) protonema

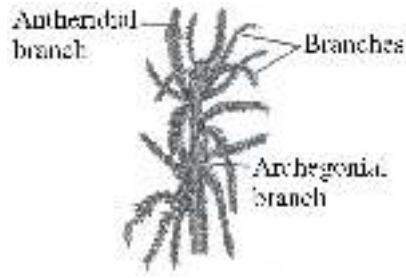
15) Chlorophyceae have :-

- (1) Chlorophyll 'a', 'b' and floridean starch
- (2) Chlorophyll 'a', 'c' and starch
- (3) Chlorophyll 'a', 'b' and starch
- (4) Chlorophyll 'a', 'c' and floridean starch

16) Which of the following is last stage of sporophytic generation in pteridophytes ?

- (1) Spore
- (2) Embryo
- (3) Zygote
- (4) Spore mother cell

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17) Identify the given figures:-

- (1) Marchantia
- (2) Funaria
- (3) Sphagnum
- (4) Riccia

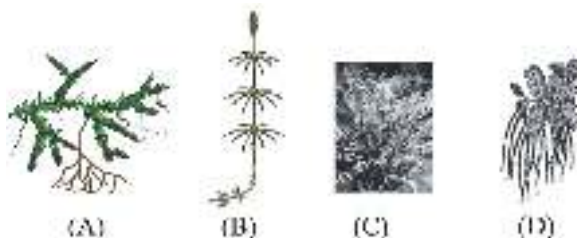
18) In which of the following the plant body is differentiated into hold fast, stipe and lamina / frond ?

- (1) *Laminaria*
- (2) *Sphagnum*
- (3) *Selaginella*
- (4) *Salvinia*



19) Identify the given diagram and select the **correct** option:-

- (1) *Ulothrix*
- (2) *Ectocarpus*
- (3) *Fucus*
- (4) *Nostoc*



20) Identify the diagrams A, B, C, D :-

- (1) A-Fern, B-Marchantia, C-Pinus, D-Fern (aquatic)
- (2) A-Riccia, B-Equisetum, C-Fern, D-Sphagnum
- (3) A-Funaria, B-Equisetum, C-Selaginella, D-Pinus
- (4) A-Selaginella, B-Equisetum, C-Fern, D-Salvinia

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(A)



(B)

21) Identify the given figures :

- (1) A - *Cycas*, B - *Ginkgo*
- (2) A - *Cycas*, B - *Cycas*
- (3) A - *Ginkgo*, B - *Cycas*
- (4) A - *Ginkgo*, B - *Ginkgo*

22) The members of rhodophyceae are commonly called red algae because of the :-

- (1) they are found in greater concentrations of warmer area of ocean.
- (2) they have floridean starch as a stored food.
- (3) predominance of r-phycoerythrin in their body.
- (4) they show complex post fertilization developments.

23) A prokaryotic autotrophic nitrogen fixing symbiont is found in :-

- (1) *Pinus*
- (2) *Cedrus*
- (3) *Cycas*
- (4) *Ginkgo*

24) **Assertion** : The members of chlorophyceae are commonly called green algae.

Reason : They have starch as stored food.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

25) **Assertion** : Vascular plants are well adapted to the terrestrial environment and dominate the land.

Reason : Vascular plants have xylem and phloem for long distance transport.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

26) **Statement-I** : In Bryophyta, Sporophyte and Gametophyte are dependent on each other.

Statement-II : In Pteridophytes, sporophyte and Gametophytes are completely independent of each

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other.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II is incorrect.

27) **Statement-I** : Majority of Pteridophytes produces two kind of spore (macro & micro) and are known as heterosporus.

Statement-II : Development of Zygotes into young embryo takes places within female gametophyte in homosporus pteridophytes.

- (1) Statement-I & Statement-II both are correct.
- (2) Statement-I is correct & Statement-II is incorrect.
- (3) Statement-I is incorrect & Statement-II is correct.
- (4) Both Statement-I & Statement-II are incorrect.

28) **Statement-I** : The Nucellus protected by envelopes and the composite structure is called an ovule.

Statement-II : The microspore mother cell is differentiated from more than one cells of Nucellus.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect.
- (3) Only Statement-I is correct.
- (4) Only Statement-II is correct.

29) Match the column and choose correct option :-

	Column-I		Column-II
(a)	Green algae	(i)	<i>Equisetum</i>
(b)	Lycopsida	(ii)	<i>Chara</i>
(c)	Brown algae	(iii)	<i>Selaginella</i>
(d)	Sphenopsida	(iv)	<i>Ectocarpous</i>

- (1) a-ii, b-iii, c-iv, d-i
- (2) a-iv, b-i, c-ii, d-iii
- (3) a-ii, b-iii, c-i, d-iv
- (4) a-iv, b-i, c-iii, d-ii

30) Cell wall materials carbohydrate are synthesized in.....

- (1) Dictyosome and sphaerosome
- (2) Golgibodies and peroxisomes
- (3) Golgibodies
- (4) Golgibodies and sphaerosomes

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31) Part of chromosomes which is related with the polarity of chromosome :-

- (1) Chromatid
- (2) Satellite
- (3) Kinetochore
- (4) Telomere

32) Which one of the following is **correct** ?

- (a) Nuclear matrix or nucleoplasm contains nucleolus and chromatin.
- (b) Outer nuclear membrane usually remains continuous with the ER and also bears ribosomes on it.
- (c) Small and less number of nucleoli are present in cells actively carrying out protein synthesis.
- (d) The nucleolus is spherical and membranous structure.

- (1) a and b
- (2) a, b and d
- (3) a, b, c and d
- (4) b, c and d

33) Proteins of cell membrane are classified on the basis of :-

- (1) Structure of protein.
- (2) Function of protein.
- (3) Lateral movement of protein
- (4) Ease of extraction

34) The quasifluid nature of _____ enable _____ movement of protein within the overall bilayer of a plasma membrane :-

- (1) Protein, lateral
- (2) Lipid, lateral
- (3) Lipid, Flip-flop
- (4) Protein, Flip-flop

35) Match the column :

	Column-I (Size)		Column-II (Organisms)
a	1-2 μ m	i	Eukaryotic cell
b	0.02-0.2 μ m	ii	PPLO
c	About 0.1 μ m	iii	Viruses
d	10-20 μ m	iv	Bacteria

- (1) a-ii, b-i, c-iii, d-iv
- (2) a-i, b-iii, c-ii, d-iv
- (3) a-iv, b-iii, c-ii, d-i
- (4) a-iv, b-ii, c-iii, d-i

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36) Unicellular organism are/have :

- I. Capable in performing the essential functions of life.
- II. Independent existence.
- III. Bacteria, amoeba, euglena and paramecium.

- (1) I & II statement incorrect
- (2) Only II & III statement correct
- (3) III statement incorrect
- (4) All statement are correct

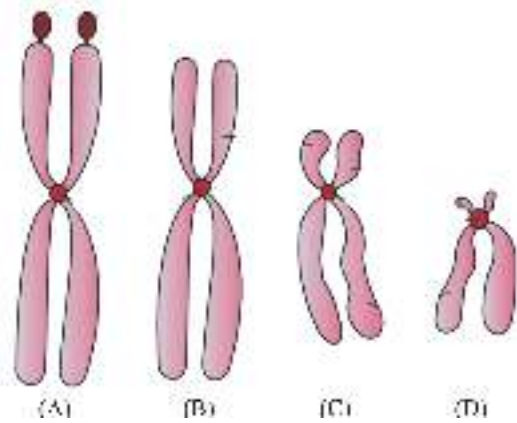
37) Which of the following statement is not correct :

- (1) Lysosome are single membrane bound structure
- (2) Lysosomes are formed by the process of packaging in golgi apparatus
- (3) Enzymes of lysosome are active at basic pH
- (4) Enzymes of lysosomes are commonly called hydrolases

38) The core of a cilium or flagellum composed of microtubule is called :-

- (1) Endomembrane system
- (2) Tubulin
- (3) Axoneme
- (4) Microfilament

39) Which of the following chromosome may represent the 13th chromosome of human genome ?



- (1) B
- (2) D
- (3) A
- (4) C

40) Select the option with correct labelling of given structure of chloroplast :-



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	A	B	C	D
(1)	Thylakoid	Granum	Stroma lamellae	Stroma
(2)	Granum	Thylakoid	Stroma lamellae	Stroma
(3)	Stroma lamellae	Thylakoid	Stroma	Granum
(4)	Stroma lamellae	Stroma	Thylakoid	Granum

(1) 1

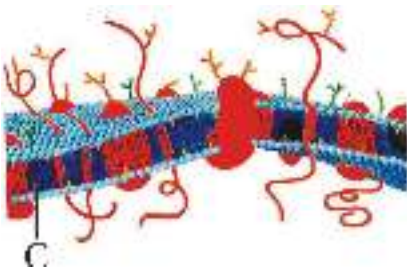
(2) 2

(3) 3

(4) 4

41)

Recognise the figure and find out the labelled C



(1) Sugar

(2) Peripheral protein

(3) Cholesterol

(4) Phospholipid

42) The chromosomal material becomes untangled during the process of chromatin condensation. This statement is **correct** for which stage :-

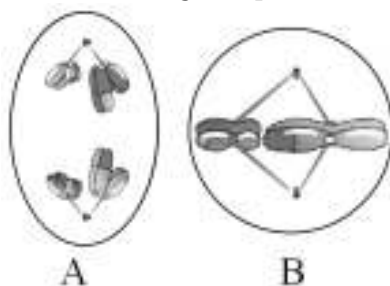
(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

43) Diagrams given below showing the phases of cell division. Choose the correct option regarding



their identification:-

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- (1) A = Anaphase-II, B = Metaphase-I
- (2) A = Anaphase-I, B = Metaphase-II
- (3) A = Anaphase-I, B = Metaphase-I
- (4) A = Anaphase-II, B = Metaphase-II

44) Chiasmata clearly observed in:

- (1) Diplotene
- (2) Pachytene
- (3) Zygotene
- (4) Leptotene

45) The microtubules from the opposite poles of the spindle attach to the pair of homologous chromosomes. This statement is correct for :

- (1) Prophase-I
- (2) Metaphase-I
- (3) Metaphase
- (4) Anaphase

46) If in a cell the amount of DNA is $2C$ in G_2 phase then what will be the ratio of amount of DNA in metaphase : G_1 phase : Anaphase ?

- (1) $2 : 1 : 2$
- (2) $1 : 2 : 2$
- (3) $2 : 1 : 4$
- (4) $1 : 1 : 2$

47) **Assertion:** Stage between two meiotic divisions is called interkinesis.

Reason: Interkinesis is short lived phase.

- (1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
- (2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
- (3) If the assertion is true but the reason is false.
- (4) If both the assertion and reason are false.

48) **Assertion :** Mitosis usually results in the production of diploid daughter cells with identical genetic complement.

Reason : The growth of multicellular organisms is due to mitosis.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

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49) **Statement-I** : Meiosis involves two sequential cycles of nuclear and cell division but only a single cycle of DNA replication.

Statement-II : Meiosis-II is initiated after the DNA replication in S-phase of interkinesis.

- (1) Both statement-I & II are correct
- (2) Both statement-I & II are incorrect
- (3) Statement-I is correct but statement-II is incorrect
- (4) Statement-I is incorrect but statement-II is correct

50) Consider the following information :-

- (a) Pyriform biflagellated male gametes
- (b) Oogamous type of sexual reproduction
- (c) Stored food - mannitol

Above informations are related to :-

- (1) *Chara*
- (2) *Fucus*
- (3) *Cladophora*
- (4) *Polysiphonia*

51) The correct sequence of events in the life cycle of gymnosperm is -

- (A) Four haploid megaspore.
- (B) Female cone on female sporophyte.
- (C) Endosporic germination.
- (D) Zygote formation.
- (E) Megaspore mother cell differentiates from one of the cell of nucellus .

Choose the correct answer from the options given below ?

- (1) A, B, C, D, E
- (2) E, B, A, C, D
- (3) A, C, B, D, E
- (4) B, E, A, C, D

52) Which one of the following matches is correct ?

1.	Cycas	Gymnosperm	Corralloid root
2.	Azolla	Pteridophyte	Mycorrhizal association
3.	Sphagnum	Gymnosperm	Peat moss
4.	Fucus	Bryophyta	Diplontic life cycle

- (1) 1
- (2) 2
- (3) 3
- (4) 4

53) How many statements are incorrect from the following statements ?

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- (i) All pteridophytes are heterosporous.
- (ii) In selaginella leaves are small (microphylls)
- (iii) Male cone are group of mega sporophylls.
- (iv) In Pinus ovule are not enclosed by ovary wall.
- (v) In fern female gametophyte are known as protonema.

Options :

- (1) Two
- (2) Four
- (3) Three
- (4) One

54) Find out the incorrect statements from the following statements ?

- (1) M-phase is the most dramatic period of the cell cycle.
- (2) In prophase spindle fibres attach to kinetochores of chromosomes.
- (3) In anaphase chromatids moves to opposite poles.
- (4) In telophase nucleolus, golgicomplex and E.R. reforms.
- (5) Liquid endosperm in coconut is an example of syncytium.

Options :-

- (1) Two
- (2) Four
- (3) One
- (4) Five

55) At the end of meiosis-II, four haploid daughter cells are formed, each genetically distinct from the parent cell. This is because of crossing over in prophase-I. Identify these stage of prophase-I, comes after which phase ?

- (1) Pachytene
- (2) Zygotene
- (3) Diplotene
- (4) Diakinesis

56) Match the following and select the correct answer ?

(a)	Centriole	(i)	Little nucleus
(b)	Mitochondria	(ii)	endoplasmic reticulum
(c)	Cisternae	(iii)	Basal body cilia or flagella
(d)	Nucleolus	(iv)	Power house of cell

- (1) a-(i), b-(iv), c-(iii), d-(ii)
- (2) a-(ii), b-(iii), c-(i), d-(ii)
- (3) a-(iii), b-(iv), c-(ii), d-(i)
- (4) a-(i), b-(ii), c-(iii), d-(iv)

57) When the centromere is situated in the middle of two equal arms of chromosomes, the

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chromosome is referred as :-

- (1) Metacentric
- (2) Telocentric
- (3) Sub-metacentric
- (4) Acrocentric

58) If a person takes deep and fast breaths for some time forcefully then after this act his desire to breath reduces for some time. This is due to:

- (1) Increased pO_2 in blood
- (2) Decreased pCO_2 in blood
- (3) Decreased pO_2 in blood
- (4) Increased pCO_2 in blood

59)

Respiratory centre is situated in :-

- (1) Cerebellum
- (2) Medulla oblongata
- (3) Hypothalamus
- (4) Cerebrum

60) Total percentage of O_2 transported by haemoglobin or RBC is :-

- (1) 3%
- (2) 97%
- (3) 49%
- (4) 25%

61) Amount of oxygen transported by one gram of haemoglobin is :-

- (1) 1.34 ml
- (2) 13.4 ml
- (3) 20 ml
- (4) 3.7 ml

62)

When oxygen supply to tissues is inadequate, the condition is :-

- (1) Dyspnoea
- (2) Hypercapnoea
- (3) Hypoxia
- (4) Apnoea

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63) In mammals the body cavity is partitioned into thoracic and abdominal parts by

- (1) Liver
- (2) Lungs
- (3) Ribs
- (4) Diaphragm

64) In the given statements choose the odd one :-

- (1) Thoracic chamber is anatomically an air-tight chamber
- (2) Any change in volume of the thoracic cavity is reflected in the lung (pulmonary) cavity
- (3) The inner pleural membrane is in close contact with thoracic lining
- (4) The inner pleural membrane is in contact with the lung surface

65) Switch off point of inspiration is _____

- (1) Aortic body
- (2) Carotid body
- (3) Pneumotaxic centre
- (4) Chemosensitive area

66) Effect of increasing 2,3-DPG on the human blood is that :-

- (1) It increases the affinity of O_2 for haemoglobin
- (2) It decreases the affinity of O_2 for haemoglobin
- (3) It increases association of Hb and O_2 .
- (4) It decreases the value of P_{50}

67) The maximum volume of air a person can breathe in after a forced expiration is :-

- (1) TV + ERV
- (2) TV + IRV
- (3) TV + ERV + IRV
- (4) ERV + RV

68) Mark the correct pair of muscles involved in the normal breathing in humans :-

- (1) Diaphragm and external intercostal muscles
- (2) Diaphragm and abdominal muscles
- (3) External and internal intercostal muscles
- (4) Diaphragm and internal intercostal muscles

69) Which of the following group of animals use lungs for respiration ?

- (1) Fish, Frog, Tadpole larva
- (2) Fish, Molluscs, aquatic arthropods

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(3) Reptiles, Birds, Mammals

(4) Insects, Earthworm, Molluscs

70) Air filled in dead space is :-

(1) 150 ml

(2) 350 ml

(3) 500 ml

(4) 1500 ml

71) A person suffers punctures in his chest cavity in an accident, without any damage to the lungs. Its effect could be :-

(1) Reduced breathing rate

(2) Rapid increase in breathing rate

(3) No change in respiration

(4) Cessation of breathing

72) Respiratory rhythm centre is located in :-

(1) Cerebellum

(2) Medulla - Oblongata

(3) Cerebral cortex

(4) Mid brain

73) Which of the following part of nephron is not present in cortex :-

(1) PCT

(2) DCT

(3) Malpighian corpuscle

(4) Collecting duct

74) Structural and functional unit of kidney is :-

(1) Neurons

(2) Nephrons

(3) Both (1) & (2)

(4) Alveoli

75) Malpighian body is :-

(1) Bowman's capsule

(2) Renal tubule

(3) Glomerulus

(4) Both (1) and (3)

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76) The cortex extends in between the medullary pyramids called :

- (1) Renal papilla
- (2) Renal column of bertini
- (3) Column of bellini
- (4) Both (1) and (2)

77) Humans are :-

- (1) Uricotelic
- (2) Ureotelic
- (3) Ammonotelic
- (4) CAD

78) Which one of following is the simplest excretory organs :-

- (1) Alveoli
- (2) Flame cells
- (3) Nephridia
- (4) Kidney

79) The weight of human kidney is :

- (1) 120-170 gm
- (2) 102-107 gm
- (3) 10.2-10.7 gm
- (4) 1020-1070 gm

80) Kidneys are retroperitoneal because :

- (1) They are anterior to peritoneum
- (2) They are posterior to peritoneum
- (3) They are lateral to peritoneum
- (4) They are latero-anterior to peritoneum

81) **Assertion** : Emphysema is a chronic disorder of Alveoli.

Reason : One of the major causes of Emphysema is cigarette smoking in which alveolar walls are damaged due to which respiratory surface is increased.

- (1) Assertion and reason both are true and the reason is correct explanation of assertion.
- (2) Assertion and reason both are true and the reason is not correct explanation of assertion.
- (3) Assertion is true but reason is wrong.
- (4) Assertion and reason both are wrong.

82) **Assertion** : Solubility of CO₂ is 20-25 times higher than that of O₂.

Reason : The amount of CO₂ that can diffuse through the diffusion membrane per unit difference in

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partial pressure is much lower as compared to that of O_2 .

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

83) Partial pressure (in mm Hg) of Oxygen and Carbon dioxide at Different parts Involved in Diffusion is given in table then identify the A, B, C and D :-

Respiratory Gases	O_2	CO_2
Atmospheric Air	159	0.3
A	104	40
Blood (Deoxygenated)	40	B
Blood (Oxygenated)	C	40
Tissues	40	D

- (1) A - Pulmonary artery, B-45, C-95, D-45
- (2) A - Pulmonary vein, B-45, C-104, D-40
- (3) A - Alveoli, B-45, C-95, D-45
- (4) A - Alveoli, B-40, C-104, D-45

84)

Match the pair :-

a $\rightarrow O_2$ (i) Carboxyhaemoglobin

b $\rightarrow CO_2$ (ii) Oxyhaemoglobin

c $\rightarrow O_3$ (iii) Carbaminohaemoglobin

d $\rightarrow CO$ (iv) Methemoglobin

- (1) a-ii, b-i, c-iii, d-iv
- (2) a-ii, b-iii, c-i, d-iv
- (3) a-ii, b-iii, c-iv, d-i
- (4) a-i, b-ii, c-iii, d-iv

85)

Match the columns and choose the correct option from the following:

Column-I		Column-II	
I	Residual volume	A	2500 ml
II	Vital capacity	B	3500 ml

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III	IRV	C	1200 ml
IV	Inspiratory capacity	D	4500 ml

- (1) II-A, III-D
- (2) III-C, IV-A
- (3) IV-B, I-C
- (4) I-D, II-B

86) Read the following statements carefully and choose the correct answer.

Statement-1 : O_2 can bind with haemoglobin in a reversible manner.

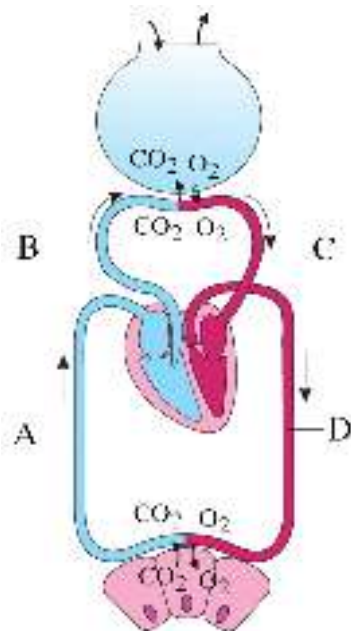
Statement-2 : CO_2 (About 20-25%) is carried by haemoglobin as carbamino-Haemoglobin.

- (1) Statement-1 and Statement-2 both are correct.
- (2) Statement-1 and Statement-2 both are incorrect.
- (3) Statement-1 is incorrect only.
- (4) Statement-2 is incorrect only.

87) **Statement - I** : P_{O_2} is the major factor that could affect the binding of CO_2 with haemoglobin.

Statement - II : The role of oxygen in the regulation of respiratory rhythm is quite insignificant. Identify option which describes above mentioned statements correctly.

- (1) Both statement - I and II are true.
- (2) Statement - I is true but II is false.
- (3) Statement - I is false but II is true.
- (4) Both statement I & II are false.



88) Study the above diagram and select the correct option regarding identification and statement :-

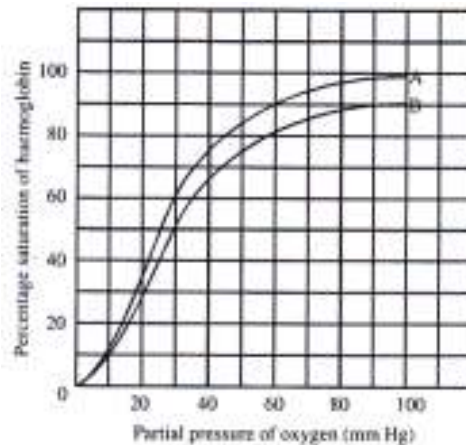
- (1) A - Systemic vein, Chloride content are more in RBC than plasma.
- (2) B - Pulmonary Artery, $PO_2 = 95-104$ mmHg,
 $PCO_2 = 40$ mmHg

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(3) C - Pulmonary vein, carry oxygenated blood upto systemic organs.

(4) A - Systemic artery, deliver 20 ml of O₂ to tissue fluid

89) Following is the oxygen-haemoglobin dissociation curve, mark the conditions in which curve will



shift from A to B.

b. Increase in partial pressure of carbon dioxide.

c. Low temperature.

d. Increase in pH, blood becomes more alkaline.

a. Increase in DPG.

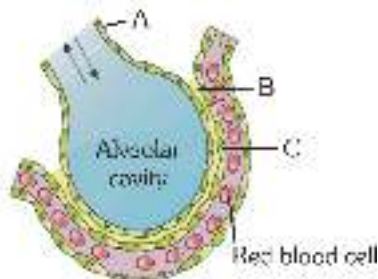
(1) a and b

(2) b and c

(3) c and d

(4) a and d

90) Find out the correct option for A, B, C in given diagram.



	A	B	C
(1)	Endothelium	Columnar cells	Squamous epithelium
(2)	Squamous epithelium	Basement membrane	Columnar cells
(3)	Squamous epithelium	Basement membrane	Endothelium
(4)	Columnar cells	Endothelium	Squamous epithelium

(1) 1

(2) 2

(3) 3

(4) 4

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ANSWER KEYS

PHYSICS

[illegible]

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	4	2	4	1	4	3	4	3	1	3	2	4	4	4	4	2	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	4	2	1	4	3	4	3	4	4	3	2	1	4	3	2	4	2	3	4
Q.	86	87	88	89	90															
A.	2	4	1	1	4															

BIOLOGY

[illegible]

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SOLUTIONS

PHYSICS

1)

Magnitude of $3\hat{i} + 4\hat{j} = 5$

Direction of $5\hat{i} + 12\hat{j} = \frac{5}{13}\hat{i} + \frac{12}{13}\hat{j}$

Required vector = $10\left(\frac{5}{13}\hat{i} + \frac{12}{13}\hat{j}\right)$

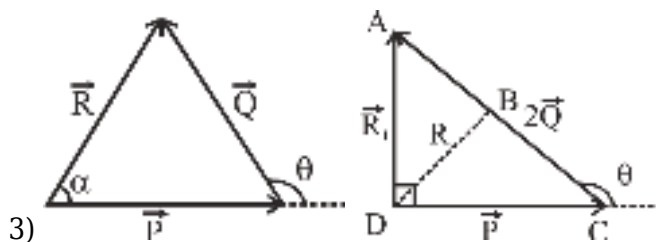
2) $(\vec{A} + \vec{B})$ is perpendicular to $(\vec{A} - \vec{B})$

Thus $(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 0$

$$\Rightarrow \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{B} = 0$$

$$\Rightarrow A^2 - B^2 = 0 \Rightarrow A = B$$

Thus the ratio of magnitudes $A/B = 1$



3)

$$\vec{P} + \vec{Q} = \vec{R}$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

When Q is doubled,

resultant is perpendicular to $\vec{P}$

$$\square R^2 = P^2 + Q^2 + 2PQ \cos \theta$$

$$R_1^2 = P^2 + 4Q^2 + 4PQ \cos \theta$$

From right angled triangle ADC

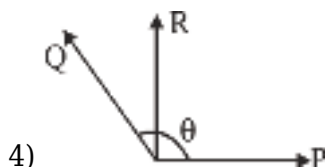
$$4Q^2 = R_1^2 + P^2$$

$$R_1^2 = 4Q^2 - P^2$$

Substituting in (ii) and solving we get

$$P^2 + 2PQ \cos \theta = 0$$

Substituting (iii) in (i), we get, $R = Q$



4)

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$$\Rightarrow \tan 90^\circ = \frac{Q \sin \theta}{P + Q \cos \theta} \Rightarrow P + Q \cos \theta = 0$$

$$\Rightarrow \cos \theta = \frac{-P}{Q} \quad \theta = \cos^{-1} \left(\frac{-P}{Q} \right)$$

5) If initial and final position is same result will be zero

Using triangle law

$$\vec{B} + \vec{C} - \vec{A} = \vec{0}$$

$$6) (\hat{i} - 3\hat{j} + 2\hat{k}) + (3\hat{i} + 6\hat{j} - 7\hat{k}) + r = \hat{j} \Rightarrow r = -4\hat{i} - 2\hat{j} + 5\hat{k}$$

$$7) R_{\max} = 8 + 6 = 14 \text{ N}$$

$$R_{\min} = 8 - 6 = 2 \text{ N}$$

$$8) \vec{A} \cdot \vec{B} = (3\hat{i} + 4\hat{j} + 5\hat{k}) \cdot (3\hat{i} + 4\hat{j} - 5\hat{k}) = 9 + 16 - 25 = 0$$

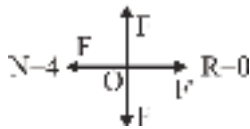
$$\theta = 90^\circ$$

$$9) C^2 = A^2 + B^2 + 2AB \cos \theta$$

$$(13)^2 = (12)^2 + (5)^2 + 2 \cdot 12 \cdot 5 \cdot \cos \theta$$

$$\cos \theta = 0 \Rightarrow \theta = 90^\circ$$

$$10) \text{ If } N = 2, \vec{F} \text{ and } \vec{O}, R = 0$$



11) Resultant of two vectors $\vec{A}$ and $\vec{B}$ can be given by $\vec{R} = \vec{A} + \vec{B}$

$$|\vec{R}| = |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\text{If } \theta = 0^\circ \text{ then } |\vec{R}| = A + B = |\vec{A}| + |\vec{B}|$$

12)

$$\text{Resultant} = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$= \sqrt{(2)^2 + (\sqrt{3})^2 + 2 \times 2 \times \sqrt{3} \cos 30^\circ}$$

$$= \sqrt{4 + 3 + 4\sqrt{3} \times \frac{\sqrt{3}}{2}}$$

$$= \sqrt{7 + 6}$$

$$= \sqrt{13}$$

13) Displacement = 0

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as initial & final position of object is same.

$$14) x = 10t$$

$$0 < t < 5 \text{ sec for } t = 5 \text{ sec,}$$

$$x_1 = 50 \text{ m}$$

$$0 = 10 - a(10 - 5) \Rightarrow a = 2 \text{ m/s}^2$$

$$x - x_1 = 10(t - 5) - \frac{1}{2} a(t - 5)^2 (5 < t < 10)$$

$$\Rightarrow x = 50 + 10t - 50 - t^2 - 25 + 10t$$

$$x = -t^2 + 20t - 25$$

$$\text{at } t = 10; x_2 = 75 \text{ m for } t > 10 \text{ sec.}$$

$$x \equiv 75 \text{ m}$$

$$15) \vec{U} = \hat{i} + 2\hat{j} \quad U \cos \theta = 1 \quad U \sin \theta = 2$$

$$x = U \cos \theta \times t = 1 \times t \Rightarrow t = x$$

$$y = (U \sin \theta) t - \frac{1}{2} g t^2$$

$$y = 2t - 5t^2 = 2x - 5x^2$$

16)

Explaining : If R is correct it explain A or not.

Concept : This question is based on Instantaneous velocity & acceleration.

Solution : Acceleration is change in velocity per unit time. One cannot tell acceleration just by knowing velocity at a one instant of time only. That is why if velocity of the particle is zero at some instant then its acceleration can be zero or non-zero at that instant. Statement of reason is one example of the asserted statement it is not the correct explanation.

Final Answer : (2)

$$17) h = 0 + \frac{1}{2} g t_1^2$$

$$\Rightarrow t_1 = \sqrt{\frac{2h}{g}}$$

$$\frac{h}{4} = 0 + \frac{1}{2} g t_2^2 \Rightarrow t_2^2 = \frac{2}{g} \times \frac{h}{4}$$

$$\Rightarrow t_2 = \sqrt{\frac{2}{g} \times \frac{h}{4}} = \frac{1}{2} \sqrt{\frac{2h}{g}} = \frac{t_1}{2}$$

18)

$$S = ut + \frac{1}{2} a t^2$$

$$-h = 4.9 \times 3 - \frac{1}{2} \times 9.8 \times (3)^2$$

$$h = 29.4 \text{ m}$$

$$19) \quad t = \sqrt{\frac{2H}{g}} \Rightarrow t \propto \sqrt{H}$$

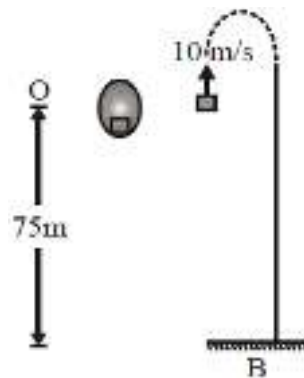
$$\frac{t_1}{t_2} = \sqrt{\frac{h}{2h}} = \frac{1}{\sqrt{2}}$$

20) $u = 0$, velocity at half of the height, $v = 10 \text{ m/s}$, $g = 10 \text{ m/s}^2$.

$$\square v^2 = u^2 + 2gs$$

$$\text{or } (10)^2 = 0 + 2 \times 10 \times \left(\frac{h}{2}\right)$$

$$\square h = 10 \text{ m}$$



21) Taking O as origin.

$$h = ut - \frac{1}{2}gt^2$$

$$-75 = 10t - 5t^2$$

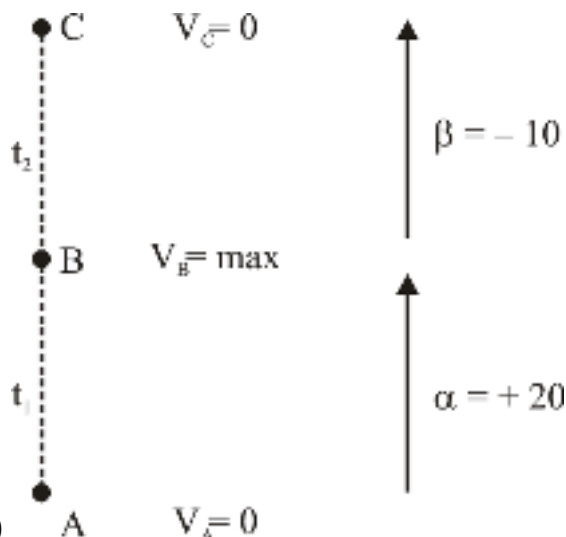
$$t^2 - 2t - 15 = 0$$

$$(t-5)(t+3) = 0$$

$$t = 5 \text{ s}$$

$$t = -3 \text{ s, not possible}$$

O to B : displacement = - 75 m



22) A $V_A = 0$

$$V_B = V_A + 20(t_1)$$

$$V_B = 0 + 100 \text{ m/s}$$

$$V_C = V_B - 10(t_2)$$

$$0 = 100 - 10(t_2)$$

$$t_2 = 10$$

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$$\begin{aligned}\text{Total dist.} &= \frac{1}{2} \frac{\alpha\beta}{(\alpha + \beta)} t^2 \\ &= \frac{1}{2} \times \frac{20 \times 10}{30} \times 15^2 \\ &= 10 \times 5 \times 15 \\ &= 750 \text{ m}\end{aligned}$$

23) Ball is coming down after $t = 2\text{s}$

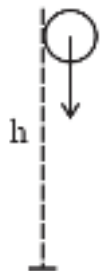
- acceleration is always downwards
- velocity is downwards as ball is coming down
- Ball is still in +y direction. So, displacement is upwards.

24)

Distance travelled in last two seconds in upward motion is same as distance travelled in first two seconds in upward motion is same as distance travelled in first two second during downward journey.

Downward journey is like free fall, so distance covered in first two seconds

$$d = \frac{1}{2} gt^2 = \frac{1}{2} (10)(2)^2 = 20 \text{ m}$$



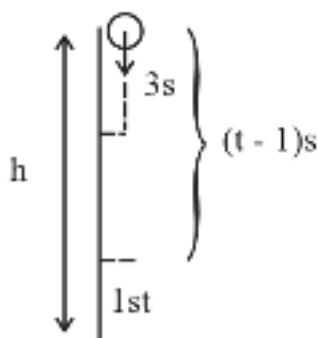
25)

$$T = \sqrt{\frac{2h}{g}} + \frac{n}{v}$$

$$h = \frac{1}{2} gt_1^2$$

$$t_1 = \sqrt{\frac{2h}{g}}$$

$$t_2 = \frac{h}{v}$$



26)

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$$\frac{1}{2}gt^2 - \frac{1}{2}g(t-1)^2 = \frac{1}{2} \times g \times (3)^2$$

$$\frac{2t-1}{2} = 9$$

$$t = 5 \text{ sec}$$

$$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (5)^2$$

$$= 125 \text{ m}$$

$$27) \vec{v} = \vec{u} + \vec{a}t$$

$$-4u = +u - gt$$

$$t = \frac{-5u}{-g} = \frac{5u}{g}.$$

28) At highest position, $v = 0$ and $a = -g$

$$29) H = -20 \times 6 + \frac{1}{2}10 \times 6 \times 6 = 60 \text{ m}$$

$$30) \frac{V_A}{V_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{3}$$

31)

$$\text{displacement} = \frac{1}{2}(3+2) \times 4 - \frac{1}{2}(2+1) \times 4$$

$$= 10 - 6 = 4 \text{ m}$$

32) Displacement = $5 - 10 = -5 \text{ m}$

33) Velocity is equal to the slope of displacement-time graph $\left(\because \frac{dy}{dx} = \frac{dx}{dt} \right)$ which is negative at the point E.

34)

Area under the graph (v-t) curve = displacement

$$\int v \cdot dt = \frac{1}{2} \times 10 \times 12 = 60 \text{ m}$$

Direction of motion is not

changing so distance = $\left| \overrightarrow{\text{disp.}} \right|$

35) From O to A, velocity will be positive.

At A, velocity is zero

From A to B, velocity is negative

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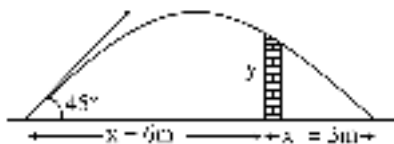
36) Maximum range $R_{\max} = \frac{u^2}{g} = \frac{(20)^2}{10} = 40\text{m}$

37) v_x will remain same while v_y will reverse
so $\vec{v}_B = 2\hat{i} - 3\hat{j}$

38) $\therefore R = \frac{U^2 \sin 2\theta}{g}, \frac{R_1}{R_2} = \frac{\sin 2\theta_1}{\sin 2\theta_2}$
 $\frac{100\sqrt{3}}{R_2} = \frac{\sin 2 \times 15^\circ}{\sin 2 \times 60^\circ}, \frac{100\sqrt{3}}{R_2} = \frac{1/2}{\sqrt{3}/2}$
 $R_2 = 300\text{ m}$

39) $\frac{u \sin \theta}{g} = t_0 = \frac{3+1}{2} = 2$
 $v_y = u \sin \theta = 2g = 20$
 $H = \left[\frac{u_y^2}{2g} \right] = \left[\frac{(20)^2}{2(10)} \right] = 20$

40) $\frac{H}{R} = \frac{\tan \theta}{4}$
 $\frac{H}{3H} = \frac{\tan \theta}{4}$
 $\tan \theta = \frac{4}{3} \Rightarrow \theta = 53^\circ$



41) $R = 6 + 3 = 9\text{m}$
 $y = (x \tan \theta) \left[1 - \frac{x}{R} \right] = 6 \times 1 \left[1 - \frac{6}{9} \right] = 2\text{m}$

42) Horizontal range
 $R = (u_x)(T)$
If projected horizontal from same height then time of flight will be same $R \propto u_x$
 $\frac{R_1}{R_2} = \frac{100}{40} = \frac{5}{2}$

43) 1. Problem Explanation

An Aeroplane moving at 180 km/h (50m / s) drops a food packet from a height of 490 m. We need to find the horizontal range of the packet before it hits the ground.

2. Concept

The packet maintains the plane's horizontal velocity and falls under gravity.

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3. Formula

- Time to fall:

$$t = \sqrt{\frac{2h}{g}}$$

- Horizontal Range:

$$R = v_x \times t$$

4. Calculation

$$t = \sqrt{\frac{2 \times 490}{9.8}} = \sqrt{100} = 10s$$

$$R = 50 \times 10 = 500 \text{ m}$$

5. Final Answer

Correct option: (3) 500 m

44)

$$R = u \sqrt{\frac{2H}{g}}$$

$$45) \vec{v}_x = 10\hat{i}$$

$$\vec{v}_y = -10\hat{j} \quad t = 1.5 \text{ sec.} \quad \vec{a} = -g\hat{j} = -10\hat{j}$$

$$(\vec{u}_y) = -10\hat{j} (-10\hat{j}) \times (1.5)$$

$$\Rightarrow \vec{u}_y = 5\hat{j}$$

$$\Rightarrow \vec{u} = 10\hat{i} + 5\hat{j}$$

$$\Rightarrow |\vec{u}| = 5\sqrt{5} \text{ m/s}$$

CHEMISTRY

46)

Solution :

Molar mass of $\text{CaCO}_3 = 40 \text{ (Ca)} + 3 \times 16 \text{ (O)} = 100 \text{ g/mol}$

Molar mass of $\text{CaO} = 40 \text{ (Ca)} + 16 \text{ (O)} = 56 \text{ g/mol}$

From 100 g CaCO_3 , we get 56 g CaO residue.

$$\text{So, from 200 g } \text{CaCO}_3, \text{ CaO residue} = \frac{56}{100} \times 200 = 112 \text{ g}$$

Final Answer :

Option (4)

47) 20g of NaOH dissolved in 100 ml of solution

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$$\text{molarity} = \frac{20}{40 \times 100} \times 100 = 5\text{M}$$

$$48) \frac{3.011 \times 10^{22}}{6.022 \times 10^{23}} = \frac{1.15}{M_w}$$

$$\boxed{M_w = 23}$$

49) NCERT Pg. # 23, Q. 1.13

50)

Molecule weight of $\text{CaCO}_3 = 100 \text{ g}$

So total number of electron in 1 mole $\text{CaCO}_3 = 50 N_A$

$$= 3.011 \times 10^{25}$$

51)

$$PV = nRT \quad 2V = 2 \times 0.0821 \times 273$$

$$v = 0.0821 \times 273$$

52)

no. of neutrons = mole $\times N_A \times$ neutron

$$= \left(\frac{18}{18} \right) \times N_A \times 8$$

$$= 8 N_A$$

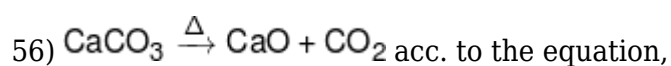
53)

$$\text{V.D.} = \frac{\text{M.wt}}{2}$$
$$\frac{(\text{V.D.})_1}{(\text{V.D.})_2} = \frac{(\text{M.wt})_1}{(\text{M.wt})_2}$$

54)

$$\% \text{ of O by mass} = \frac{\text{At. wt. of "O"} \times \text{Atomicity}}{M_w \text{ of Nylon}}$$

55) Equal moles of different substances contain same number of constituent particles. But equal weights of different substances do not contain the same number of constituent particles. Hence, Assertion is correct but reason is incorrect.



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100 g CaCO_3 gives 22.4 L of CO_2
 $\therefore$ 10 g CaCO_3 will give 2.24 L of CO_2
When CaCO_3 is 100% pure
 $\therefore$ 100% $\rightarrow$ 2.24 L
90% $\rightarrow$ 2.016 L

57)

$\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$; Pressure increases, reaction move towards Lesser number of gaseous moles.

58) $P \uparrow$ volume $\downarrow$
 $\Rightarrow$ Reaction goes in forward direction

59) Le-Chatlier's principle

60)

On decreasing volume, molarity of each substance increases (if $\Delta n_g = 0$)

61)

Theory based

62)

Explain Question : For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{Heat}$, the equilibrium shifts in the forward direction under which condition.

Concept : This question is based on Le Chatelier's Principle

Solution :

Option 1: Increasing NH_3 (product) $\rightarrow$ shifts backward

Option 2: Decreasing pressure $\rightarrow$ shifts to **more gas molecules (backward)**

Option 3: Decreasing N_2 & $\text{H}_2 \rightarrow$ fewer reactants $\rightarrow$ **shifts backward**

Option 4: Increasing pressure (Favors 4 $\rightarrow$ 2 moles) & decreasing temperature (Favors exothermic) $\rightarrow$ **shifts forward**

Final Answer : option (4)

63) Anions undergo hydrolysis
Hence $\rightarrow$ Alkaline

64) NCERT Pg. # 225

AIM: FREE EDUCATION TO ALL APUL

65) NCERT Pg. # 225

$$66) \text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_2\text{OH}]} = 6$$
$$\square \text{pH} = 14 - 6 = 8.0$$

67) **Explanation:**

Find buffer solution

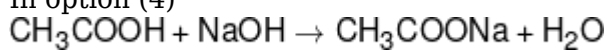
Concept:

This concept based on Buffer Solution & P^{H}

Solution:

Acidic Buffer is a Solution of W.A & conjugate salt with strong Base.

In option (4)



$$.5 \times .2 \quad .5 \times .1 \quad 0 \quad -$$

$$.1 \quad .05 \quad 0 \quad -$$

$$.05 \quad 0 \quad .05$$

Final Solution have CH_3COOH & CH_3COONa both So, 'H' is a buffer solution.

Correct option:

(4) 500 mL of 0.2 N CH_3COOH + 500mL to 0.1 N NaOH

$$68) [\text{OH}^-] = 2 \times 10^{-3}$$

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = 3 - \log 2 = 2.7$$

$$\text{then pH} = 14 - 2.7 = 11.3$$

69) Acidic buffer is prepared by mixing weak acid and its conjugate base.

70) Since $K_b > K_a$, the solution will be slightly basic.

71) Weak acid [HA]

$$[\text{H}^+] = c\alpha$$

$$\text{and } [\text{H}^+] [\text{OH}^-] = K_w \Rightarrow [\text{OH}^-] = \frac{K_w}{[\text{H}^+]}$$

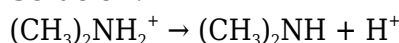
72) **Question Explanation :**

Question is asking about conjugated base of the given acid.

Concept :

This question is based on Bronsted Lowry Concept.;

Solution:



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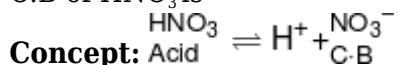
Final Answer :

Option (4)

73) Solution/Explanation/Calculation:

Explanation:

C.B of HNO_3 is-



Final Answer = NO_3^-

74) Question Explanation :

We are asked to identify the salts that undergo hydrolysis in water.

Concept :

This question is based on Salt hydrolysis.

Solution :

1. Na_3PO_4 : This salt is formed from a strong base (NaOH) and a weak acid (H_3PO_4). The PO_4^{3-} ion can accept protons from water, making the solution basic. So, Na_3PO_4 undergoes hydrolysis.

2. CH_3COONa : This salt is formed from a strong base (NaOH) and a weak acid (CH_3COOH). The CH_3COO^- ion can accept protons from water, making the solution basic. So, CH_3COONa undergoes hydrolysis.

3. NaNO_3 : This salt is formed from a strong base (NaOH) and a strong acid (HNO_3). Neither ion will react significantly with water, so NaNO_3 does not undergo hydrolysis.

Final Answer:

The correct option is (4).

75) $[\text{H}^+]_1 = 10^{-2}$; $[\text{H}^+]_2 = 10^{-4}$

$$\frac{[\text{H}^+]_2}{[\text{H}^+]_1} = \frac{10^{-4}}{10^{-2}} = 10^{-2}$$

76) SBWA

77) Explanation:

Most Acidic Solution

Concept:

This concept based on P^{H} Calculation

Solution:

we now

Acidic strength $\propto \frac{1}{\text{P}^{\text{H}}}$

Solⁿ "B" show least value of P^{H} so it is most Acidic Solution.

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Correct option : (2)

$$78) [H^+] = \sqrt{K_a \times C}$$

79)

Hydrochloric acid (HCl) is a strong electrolyte its degree of ionisation(α) is 100% whereas acetic acid (CH₃COOH) is a weak electrolyte its $\alpha \ll \ll 100\%$

$$80) \alpha = \sqrt{\frac{K_a}{C}}$$

$$\alpha = \sqrt{\frac{10^{-5}}{10^{-1}}} = 10^{-2}$$

$$\% = \alpha \times 100$$

$$= 10^{-2} \times 100$$

$$= 1\%$$

81)

Species which releases H⁺ ions are Acid is species.

82)

Question

The question asks us to determine the volume of water needed to be added to a 1-liter HCl solution with pH 1 to dilute it to a solution with pH 2.

Given data

- A. Initial volume (V₁) : 1 liter
- B. Initial pH (pH₁): 1
- C. Final pH (pH₂): 2

Concept:

- A. pH and [H⁺] Relationship : $pH = -\log[H^+] = 10^{-pH}$

Calculation

- 1. Calculate initial [H⁺] (M₁):

$$A. [H^+]_1 = 10^{(-pH_1)} = 10^{(-1)} = 0.1M$$

- 2. Calculate final [H⁺] (M₂):

$$A. [H^+]_2 = 10^{(-pH_2)} = 10^{(-2)} = 0.01M$$

- 3. Use the dilution formula (M₁V₁=M₂V₂) to find the final volume (V₂).

$$A. (0.1 M) (1 L) = (0.01M) (V_2)$$

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B. $V_2 = (0.1 \text{ M} \times 1\text{L}) / 0.01 \text{ M}$

C. $V_2 = 10\text{L}$

4. Calculate the volume of water added:

A. Volume of water added = $V_2 - V_1$

B. Volume of water added = $10\text{L} - 1 \text{ L} = 9\text{L}$

Conclusion

Hence, option (4) is correct.

83)

$$\text{pH} = 10 \Rightarrow [\text{H}^+] = 10^{-10}$$

$$[\text{OH}^-] = \frac{10^{-14}}{10^{-10}} = 10^{-4}$$

84) Question Asking About:

A conjugate base is formed when an acid donates a proton (H^+). The species left after losing H^+ is the conjugate base

Key Concept:

HSO_4^- (bisulfite ion) acts as an acid and donates a proton (H^+).

Conclusion:

Hence, option (3) is correct.

85) The conjugate based of weak acid is strong.

86) At infinite dilution weak electrolyte behave as strong electrolyte.

87)

Half neutralised means

$$[\text{NH}_4\text{OH}] = [\text{NH}_4\text{Cl}]$$

$$[\text{OH}^-] = K_b \times \frac{[\text{NH}_4\text{OH}]}{[\text{NH}_4\text{Cl}]}$$

$$[\text{OH}^-] = K_b$$

$$[\text{OH}^-] = 2 \times 10^{-4}$$

$$\text{pOH} = 4 - \log 2 = 4 - 0.3 = 3.7$$

$$\text{pH} = 14 - 3.7 = 10.3$$

88)

$$\text{m. mole of } \text{H}^+ = 100 \times 0.2 \times 2 = 40$$

$$\text{m. mole of } \text{OH}^- = 100 \times 0.2 = 20$$

So, solution will be acidic

89) Question Explanation -

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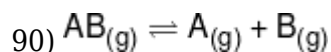
To determine pH of 10^{-5} M HCl

Concept -

$$[H^+] = 10^{-5}$$

$$\text{pH} = 5$$

Final Answer : Option (1)



Initial moles 1 0 0

Moles at eqm. 1-0.33 0.33 0.33

Total moles at eqm. = 1.33

$$P_{AB} = \left(\frac{0.67}{1.33} \right) \cdot P, P_A = \left(\frac{0.33}{1.33} \right) \cdot P, P_B = \left(\frac{0.33}{1.33} \right) \cdot P$$

$$K_p = \frac{P_A \times P_B}{P_{AB}} = \frac{\left(\frac{0.33}{1.33} \cdot P \right) \times \left(\frac{0.33}{1.33} \cdot P \right)}{\left(\frac{0.67}{1.33} \cdot P \right)}$$

$$= \frac{0.33 \times 0.33 \times P}{0.67 \times 1.33} = 0.122 P$$

$$K_p = 0.122 P \quad P = \frac{K_p}{0.122} = 8K_p$$

BIOLOGY

91)

NCERT (XI) Pg. # 26

92) NCERT (XIth) Pg. # 26

93) NCERT (XIth) Pg. # 27

94) NCERT XI, Pg.# 28

95)

NCERT-XI, Pg. # 27

Explain Question : Question asking about red algae

Concept : This question is based on Red algae

Solution : all statements are given in question, related to red algae (Rhodophyceae)

Reference NCERT/11/TH/Page 27

Rhodophyceae

The members of rhodophyceae are commonly called red algae because of the predominance of the red pigment, r-phycoerythrin in their body. Majority of the red algae are marine with

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greater concentrations found in the warmer areas. They occur in both well-lighted regions close to the surface of water and also at great depths in oceans where relatively little light penetrates. The red thalli of most of the red algae are multicellular. Some of them have complex body organisation. The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure. The red algae usually reproduce vegetatively by fragmentation. They reproduce asexually by non-motile spores and sexually by non-motile gametes. Sexual reproduction is oogamous and accompanied by complex post fertilisation developments.

Final Answer : option (3)

96) NCERT-XI, Pg. # 26

Explain Question: Question asking about examples of chlorophyceae

Concept: This question is based on Algae-classification..

Solution: Volvox, Spirogyra, Chara belongs to Chlorophyceae (green algae) class in Algae.

Final Answer: (2)

97) NCERT-XI, Pg. # 33

98) NCERT-XI, Pg. # 30, 32

99) NCERT-XI, Pg. 32

100) NCERT-XI, Pg. # 32

101)

NCERT-XI, Pg. # 36, 3.2.2 section
Foot and Capsule

102) NCERT XI Pg. # 26

103) NCERT-XI, Pg. # 28

Explain Question:

Question asking about to identify the given figure and select the correct matching option.



Concept:

This question is based on Bryophyta-Moss.

Solution:

The given figure is of Funaria (mosses) and this is related to class Bryopsida.

Final answer: 4

104) NCERT XI, Pg. # 29

105) NCERT-XI, Pg # 26

106) NCERT-XI Pg # 32

107)

NCERT-XI, Pg. # 28

The image shows the structure of **Sphagnum**.

108)

NCERT XI Pg.# 25

109)

NCERT-XI, Pg. # 25

Explain Question : The question is asking you to: Look at a diagram and identify which organism it represent from the given options.



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Concept : This question is based on Green algae

Solution : The correct answer is 1 - Ulothrix

Ulothrix is filamentous green alga found in freshwater.

It has:

- Unbranched filaments-Chloroplasts in girdle/bracelet shape
- Cell wall and basal holdfast

Final Answer : option (1) *Ulothrix*

110) NCERT-XI, Pg. # 31

111)

NCERT XI Pg. # 33

112) NCERT-XI, Pg. # 27

113) NCERT XI Pg. # 33

114) NCERT-XI, Pg. # 26

Explanation :

Starch → Assertion and Reason

Concept :

Polysaccharide

Solution :

Starch gives I_2 test blue colour.

→ Starch form helical secondary structure.

Final Answer :

Option (1)

115) NCERT-XI, Pg. # 33

116) NCERT (XIth) Pg. # 28, 29

117) NCERT (XIth) Pg. # 32

118) NCERT Pg. # 33

119)

NCERT-XI, Pg. # 27, 32

120) **Explaining the Question** - Question asking about the function of golgi body

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Concept - This Question is based on Golgi complex

Solution - Synthesis of some cell wall materials and Cell plate formation during cytokinesis by golgi complex

Final Answer - 3

121)

NCERT-XI, Pg. # 96

Concept:

- A. Telomeres are repetitive DNA sequences at the ends of chromosomes that protect them from damage.
- B. Telomeres shorten with each cell division, and this shortening is linked to aging.

Solution:

- A. Telomere shortening is a well-established biological marker of aging.

122) NCERT XI Pg # 100

123) NCERT-XI Pg. # 93

124) NCERT (XI) Pg. # 93

125)

NCERT-XI, Pg. # 90

Solution:

match the sizes in Column-I with the organisms in Column-II

- (a) 1-2 μ m (iv) Bacteria
- (b) 0.02-0.2 μ m (iii) Viruses
- (c) About 0.1 μ m (ii) PPLO
- (d) 10-20 μ m (i) Eukaryotic cell

Final Answer : option (3). a-iv, b-iii, c-ii, d-i

126)

NCERT-XI, Pg. # 87

Solution:

- I. Capable in performing the essential functions of life.
- II. Independent existence.
- III. Bacteria, amoeba, euglena and paramecium.

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Final Answer : option (4).
All statements are correct.

127) NCERT-XI, Pg. # 96

128) NCERT-XI, Pg # 99

129) NCERT-XI, Pg. # 101

130) NCERT XI Pg # 98

131) NCERT-XI, Pg. # 93

132) NCERT-XI, Pg. # 122

Question Explanation :

The question is asking during which stage of mitosis does the chromatin (loose DNA) condense and become untangled, forming visible chromosomes ready for proper segregation.

Concept :

Cell cycle and cell division : Mitosis.

Solution :

In prophase, the chromatin fibers condense into discrete, visible chromosomes. During this condensation, the chromosomal material untangles and organizes, which is essential for accurate chromosome separation later in mitosis.

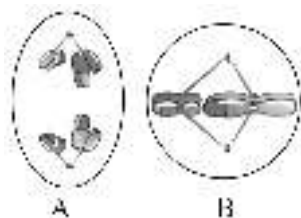
Final Answer:

Option (1)

133) NCERT-XI, Pg. # 127

Question Explanation :

The question is asking you to identify the correct meiotic phases. shown in two given diagrams labeled A and B. You must choose the option that correctly matches each diagram to its cell



division stage.

Cell cycle and cell division : Meiosis

Concept :

Solution :

The correct answer is Option 2. A = Anaphase-I, B = Metaphase-II

Explanation :

Diagram A shows Anaphase - I, homologous chromosomes separate.

Diagram B shows Metaphase-II, individual chromosomes align at the equator → single

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chromosomes at the center, so it's Metaphase-II.

Final Answer :

Option (2)

134) NCERT-XI, Pg. # 126

Question Explanation :

The question asks : During which substage of prophase I in meiosis are chiasmata (cross-shaped structures due to crossing over) clearly visible under a microscope?

Concept :

Cell cycle and cell division : Meiosis

Solution :

The correct answer is 1. Diplotene

Explanation :

Chiasmata are the physical sites where crossing over occurred between homologous chromosomes.

These become clearly visible in the diplotene stage, as homologous chromosomes start to separate, but stay connected at chiasmata.

Final Answer :

Option (1)

135) NCERT-XI, Pg. # 126

136) NCERT-XI, Pg. # 121

G2 - 2C, Metaphase is 2C: G1 is 1C: Anaphase is 2C.

137) NCERT-XI, Pg. # 127

SOLUTION:

Assertion: The stage between two meiotic divisions is called interkinesis - this is true. Interkinesis is a resting phase that occurs between meiosis I and meiosis II, and it prepares the cell for the second meiotic division.

Reason: Interkinesis is a short-lived phase - this is also true. Interkinesis does not involve DNA replication and is generally a brief stage, with a duration much shorter than the other stages of meiosis

Correct answer: 2. If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

138) NCERT-XI, Pg. # 125

139)

NCERT XI, Pg. # 125

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140) NCERT-XI, Pg. # 26, 27

141) NCERT-XI, Pg. # 32,33

142) NCERT-XI, Pg. # 28, 30, 33

143) NCERT-XI, Pg. # 31, 32, 33
Statement-(i), (iii) and (v) are incorrect.

144) NCERT-XI, Pg. # 121-125
Statement-(3) is incorrect (in metaphase) spindle fibre attaches with kinetochore)

145) NCERT-XI, Pg. # 126
This stage is Pachytene, which comes after Zygotene

146) NCERT-XI, Pg. # 95, 97, 99, 100

147) NCERT-XI, Pg. # 101

148)

NCERT-XI, Pg. # 186

149)

Explain Question : Respiratory centre is situated in ?

Concept : This question is based on Regulation of Respiration.

Solution : Respiration centre is situated in Medulla oblongata.

Final Answer : option (2)

150)

Explain Question : Total % of O_2 transported by 'Hb' or RBC is ?

Concept : This question is based on Transport of oxygen.

Solution : 97% of O_2 is Transported by Hb in the form of HbO_2

Final Answer : option (2)

151)

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Explanation :

Each gram of hemoglobin (Hb) can bind to 1.34 ml of oxygen (O₂).

In a normal healthy adult, there are about 15 grams of Hb per 100 ml of blood.

So, the total oxygen-carrying capacity of blood is:

$$1.34 \text{ ml/g} \times 15 \text{ g/100 ml} = \sim 20.1 \text{ ml O}_2 \text{ per 100 ml of blood}$$

Final Answer : (1)

152)

Decrease of O₂ in tissue is known as hypoxia.

153) **Explanation :** The diaphragm is a dome-shaped muscle that separates the thoracic cavity (containing the heart and lungs) from the abdominal cavity (containing digestive organs, etc.). It plays a crucial role in respiration by contracting and flattening during inhalation, creating space for the lungs to expand.

The correct answer is (4) : Diaphragm

154) **Explanation:**

To identify the statement that is different or does not fit with the other statements regarding the thoracic cavity and pleural membranes.

Solution:

The inner pleural membrane is in close contact with thoracic lining: This is incorrect. The outer pleural membrane also known as the parietal pleura lines the thoracic. The inner pleural membrane. (visceral pleura) is in contact with the lung surface, and the outer pleural membrane (parietal pleura) is in contact with the thoracic lining. There the inner and outer pleural membrane.

Conclusion:

This answer is option (3): The inner pleural membrane is in close contact with thoracic lining

155) NCERT, Pg No- 25

156)

2-3, DPG is product of glycolysis.

157) **Explaining :**

Asking about "definition of vital capacity

Concept :

This question is based on Respiratory volume and capacity

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Solution: The maximum volume of air a person can breathe in after a forced expiration is called as Vital capacity, It is equal to $VC = TV + ERV + IRV$

Hence,

Option (3) is correct : $TV + ERV + IRV$

158)

NCERT-XI, Pg. # 186

159)

Lungs are the respiratory organs of terrestrial animals.

160)

Volume of air in conduction zone.

161)

Thoracic cavity is air tight.

162)

Explain Question : Where is the respiratory rhythm center located?

Concept : This question is based on **Regulation of respiration**

Solution : Respiratory rhythm center is in the medulla oblongata, not cerebellum, cerebrum, cerebral cortex, or midbrain.

Final Answer : option (2)

163)

NCERT-XI, Pg. # 208

164)

NCERT-XI, Pg. # 207

165)

NCERT-XI, Pg. # 208

166)

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NCERT-XI, Pg. # 207

167)

NCERT-XI, Pg. # 205

168)

NCERT-XI, Pg. # 206

169)

NCERT-XI, Pg. # 206

170)

NCERT-XI, Pg. # 206

171)

NCERT-XI, Pg. # 190

172) NCERT Pg#189

173) A. Question: Identify A, B, C, and D based on the partial pressures of oxygen and carbon dioxide at different locations involved in gas exchange.

B. Given Data:

Respiratory Gases	O₂ (mm Hg)	CO₂ (mm Hg)
Atmospheric Air	159	0.3
A	104	40
Blood (Deoxygenated)	40	B
Blood (Oxygenated)	C	40
Tissues	40	D

Export to SheetsC. Concept: Gas exchange in the lungs and tissues, partial pressures of oxygen and carbon dioxide.D. Explanation:

- A. **A:** The location with a partial pressure of oxygen (O₂) at 104 mm Hg and carbon dioxide (CO₂) at 40 mm Hg is the alveoli. This is where gas exchange occurs between the air and the blood.
- B. **B:** Deoxygenated blood arriving at the lungs has a low partial pressure of O₂ (40 mm Hg) and a relatively high partial pressure of CO₂. The value of B, the partial pressure of CO₂ in

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deoxygenated blood, is approximately 45 mm Hg.

- C. **C:** Oxygenated blood leaving the lungs has a high partial pressure of O_2 . This value, C, is around 95 mm Hg. The CO_2 pressure remains at 40 mm Hg.
- D. **D:** Tissues have a low partial pressure of O_2 (40 mm Hg) because they are constantly consuming oxygen. They also have a higher partial pressure of CO_2 because they produce CO_2 as a byproduct of metabolism. The partial pressure of CO_2 in tissues, D, is approximately 45 mm Hg.

174) NCERT (XIth) Pg. # 274, Para- 2-3 & Module-7

175)

NCERT-XI, Pg. # 187

176) NCERT, Pg. # 274

177) NCERT, Pg. # 274 (17.4.2), 275(17.5)

178)

NCERT-XI, Pg. # 188. Fig. 14.3

179)

NCERT-XI, Pg. # 189, Fig. 14.5

180)

NCERT XI Pg # 273